

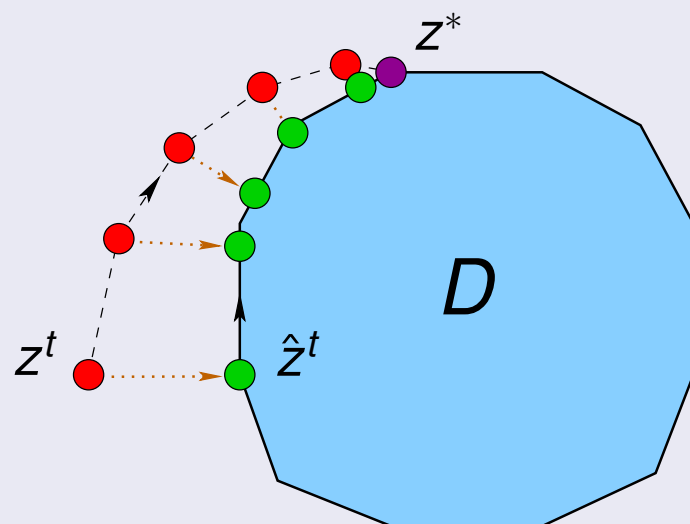


Getting Feasible Variable Estimates From Infeasible Ones for Separable Problems

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Motivation



In dual/saddle point optimization scenarios primal/primal-dual estimates are infeasible except in the optimum.

Feasibility $\Rightarrow$ valid upper/lower bound and hence:

- sound stopping condition
- adaptive optimization schemes (bundle, smoothing)
- fair comparison of diverse algorithms

Euclidean projection expensive for complex D !

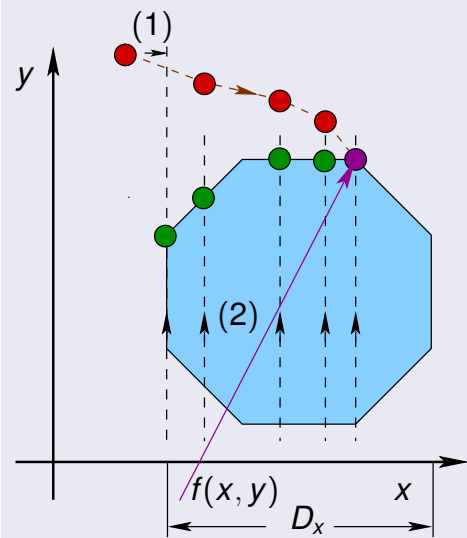
Idea

$f(z^t) \xrightarrow{t \rightarrow \infty} f^* = \arg \min_{z \in D} f(z)$ - continuous; $D \subset \mathbb{R}^n$ - closed; $z^t \equiv (x^t, y^t)$

Let $\begin{cases} \hat{x}^t := \Pi_{D_X}(x^t), & D_X = \{x \mid \exists y: (x, y) \in D\} \\ \hat{y}^t := \arg \min_{y: (\hat{x}^t, y) \in D} f(\hat{x}^t, y) \end{cases}$ # feasibility, in simple case $\hat{x}^t := x^t$
optimization w.r.t. y for a fixed x

Then $f(\hat{x}^t, \hat{y}^t) \xrightarrow{t \rightarrow \infty} f^*$ # under mild technical conditions

Linear Programming Case



$$\min_{\substack{x \in \mathbb{R}_+^{nI} \\ y \in \mathbb{R}_+^{mJ}}} \sum_{i \in I} \langle a_i, x_i \rangle + \sum_{j \in J} \langle b_j, y_j \rangle$$

$$A_{ij}y_j = x_i, \quad i \in nb(j), \quad j \in J,$$

$$B_i x_i = c_i, \quad i \in I,$$

A_{ij}, B_i - have full rank.

$$\hat{x}_i := \Pi_{\{B_i x_i = c_i, x_i \geq 0\}}(x_i) \quad (1)$$

$$\begin{cases} \hat{y}_j := \arg \min_{y_j \in \mathbb{R}_+^m} \langle b_j, y_j \rangle \\ \text{s.t. } A_{ij}y_j = \hat{x}_i \end{cases} \quad (2)$$

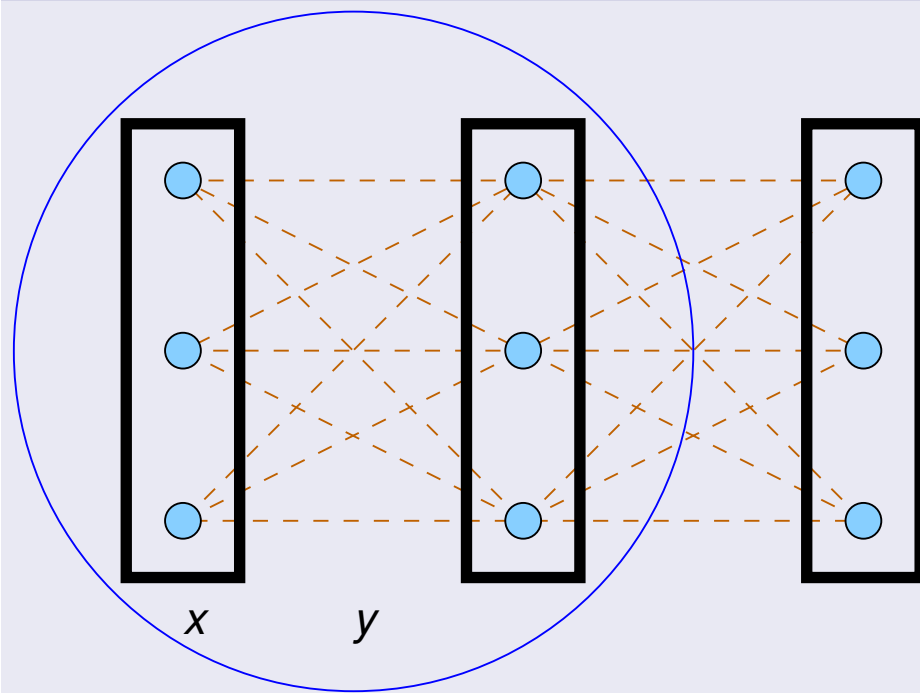
Small-sized subproblems. Good if $|I|, |J| \gg 1$

Convergence Speed

$$|f(\hat{x}, \hat{y}) - f^*| \leq \underbrace{|f(x, y) - f^*|}_{\rightarrow 0} + \underbrace{(L_Y(f) + L_X(f))}_{\geq 0} \cdot \underbrace{\|z - \Pi_D(z)\|}_{\text{Distance to } D \rightarrow 0}$$

Compare to the Euclidean projection: $|f(\hat{x}, \hat{y}) - f^*| \leq |f(x, y) - f^*| + L_{XY}(f) \|z - \Pi_D(z)\|$

Application : MRF Local Polytope Inference



eq. (2) = an independent subproblem for each p/w factor

- subproblem = a small transportation problem
- 0.01-0.02 sec for all 256×256 subproblems, each 4×4 labels at most (uniformly distributed potentials; on each iteration of a dual optimization scheme)

References

📄 Savchynskyy, B. and Schmidt, S. (2012).

Getting feasible variable estimates from infeasible ones: MRF local polytope study.

Technical report, arXiv:1210.4081.